

## RICHMOND, Australia

WMO: 943410

Lat: 20.701S

Lon: 143.120E

Elev: 207

StdP: 98.86

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>5.7 | (c)<br>7.3 | (d)<br>-10.5                  | (e)<br>1.6 | (f)<br>18.0 | (g)<br>-8.7 | (h)<br>1.8 | (i)<br>18.3 | (j)<br>10.0           | (k)<br>19.8 | (l)<br>9.2 | (m)<br>20.1 | (n)<br>2.9            | (o)<br>130 | (p)<br>0.459 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 12            | 13.7                   | 40.6            | 20.5 | 39.5 | 20.4 | 38.4 | 20.4 | 25.4                | 31.6 | 24.9 | 31.2 | 24.4 | 30.8 | 4.8                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.1                            | 19.4 | 27.3 | 23.4 | 18.7 | 27.1 | 22.9 | 18.0 | 26.9 | 79.4          | 31.5 | 77.1 | 31.4 | 75.1 | 30.4 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.4              | 9.0  | 8.0 | 2.1                        | 42.6 | 1.3                | 1.4 | 1.2                                                | 43.7 | 0.4        | 44.5 | -0.3       | 45.3 | -1.2       | 46.3 |
|                   |      |     | -0.7                       | 28.0 | 1.2                | 1.9 | -1.6                                               | 29.4 | -2.3       | 30.5 | -3.0       | 31.6 | -3.9       | 33.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.2   | 29.9  | 29.3  | 28.1  | 25.5  | 21.5  | 18.4  | 18.2  | 19.4  | 23.9  | 27.4  | 29.7  | 30.8  |  |  |
|                                                                        | DBStd       | 5.34   | 2.39  | 2.23  | 2.12  | 2.42  | 3.03  | 3.46  | 3.22  | 3.20  | 3.06  | 2.95  | 2.37  | 2.30  |  |  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 122    | 0     | 0     | 0     | 1     | 8     | 42    | 43    | 26    | 2     | 1     | 0     | 0     |  |  |
|                                                                        | CDD10.0     | 5534   | 618   | 541   | 561   | 465   | 358   | 253   | 253   | 292   | 416   | 540   | 592   | 646   |  |  |
|                                                                        | CDD18.3     | 2616   | 360   | 308   | 303   | 216   | 108   | 45    | 38    | 60    | 168   | 282   | 342   | 387   |  |  |
|                                                                        | CDH23.3     | 35249  | 4794  | 4003  | 3792  | 2577  | 1211  | 533   | 563   | 978   | 2297  | 3967  | 4909  | 5624  |  |  |
|                                                                        | CDH26.7     | 19628  | 2792  | 2267  | 2081  | 1302  | 433   | 128   | 138   | 353   | 1202  | 2361  | 3042  | 3531  |  |  |
| (14) Wind                                                              | WSAvg       | 4.2    | 4.0   | 4.0   | 4.1   | 4.3   | 4.3   | 4.2   | 4.0   | 4.2   | 4.2   | 4.5   | 4.4   | 4.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 489    | 128   | 103   | 59    | 23    | 14    | 11    | 8     | 5     | 6     | 17    | 31    | 77    |  |  |
|                                                                        | PrecMax     | 1054   | 664   | 281   | 304   | 206   | 69    | 107   | 77    | 70    | 58    | 132   | 204   | 248   |  |  |
|                                                                        | PrecMin     | 201    | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 8     |  |  |
|                                                                        | PrecStd     | 197    | 124   | 71    | 67    | 42    | 20    | 23    | 18    | 13    | 13    | 28    | 37    | 57    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 42.0  | 40.7  | 38.8  | 36.3  | 33.8  | 31.1  | 31.6  | 34.1  | 37.6  | 40.2  | 41.1  | 41.9  |  |  |
|                                                                        |             | MCWB   | 22.1  | 21.0  | 20.9  | 20.1  | 19.2  | 17.8  | 18.0  | 17.2  | 18.3  | 19.2  | 20.2  | 20.4  |  |  |
|                                                                        | 2%          | DB     | 39.9  | 38.6  | 37.2  | 35.1  | 31.9  | 29.5  | 29.6  | 32.1  | 36.0  | 38.7  | 39.7  | 40.5  |  |  |
|                                                                        |             | MCWB   | 21.5  | 21.5  | 21.0  | 19.6  | 18.5  | 17.5  | 16.9  | 16.4  | 17.6  | 18.9  | 19.8  | 20.5  |  |  |
|                                                                        | 5%          | DB     | 38.1  | 37.1  | 36.0  | 34.0  | 30.7  | 28.2  | 28.1  | 30.2  | 34.4  | 37.3  | 38.6  | 39.4  |  |  |
|                                                                        |             | MCWB   | 21.9  | 21.6  | 20.7  | 19.2  | 17.8  | 16.8  | 15.9  | 15.8  | 17.2  | 18.5  | 19.6  | 20.5  |  |  |
|                                                                        | 10%         | DB     | 36.6  | 35.9  | 34.8  | 32.9  | 29.2  | 26.8  | 26.7  | 28.6  | 33.0  | 36.0  | 37.4  | 38.1  |  |  |
|                                                                        |             | MCWB   | 21.9  | 21.7  | 20.9  | 19.1  | 17.4  | 16.0  | 15.1  | 15.0  | 16.7  | 18.3  | 19.5  | 20.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.2  | 26.0  | 25.7  | 23.8  | 21.7  | 19.8  | 20.3  | 19.6  | 21.4  | 22.5  | 24.1  | 25.2  |  |  |
|                                                                        |             | MCDB   | 33.0  | 32.0  | 30.4  | 30.5  | 29.4  | 27.4  | 27.1  | 29.2  | 30.6  | 30.3  | 32.6  | 32.1  |  |  |
|                                                                        | 2%          | WB     | 25.4  | 25.2  | 24.9  | 22.6  | 20.5  | 18.9  | 18.7  | 18.3  | 20.2  | 21.2  | 23.1  | 24.5  |  |  |
|                                                                        |             | MCDB   | 32.2  | 31.4  | 29.9  | 29.7  | 28.0  | 26.6  | 26.6  | 28.0  | 30.1  | 31.5  | 32.0  | 32.0  |  |  |
|                                                                        | 5%          | WB     | 24.8  | 24.8  | 24.2  | 21.7  | 19.5  | 17.9  | 17.5  | 17.1  | 19.2  | 20.5  | 22.4  | 23.9  |  |  |
|                                                                        |             | MCDB   | 31.6  | 30.9  | 29.6  | 29.3  | 27.1  | 26.0  | 25.5  | 27.2  | 29.9  | 31.9  | 31.6  | 31.8  |  |  |
|                                                                        | 10%         | WB     | 24.3  | 24.3  | 23.4  | 20.7  | 18.6  | 17.0  | 16.2  | 16.0  | 18.0  | 19.7  | 21.7  | 23.3  |  |  |
|                                                                        |             | MCDB   | 30.7  | 30.3  | 29.4  | 29.1  | 26.7  | 24.8  | 24.6  | 26.2  | 29.7  | 31.8  | 31.3  | 31.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.9  | 12.2  | 13.1  | 14.5  | 15.0  | 15.4  | 16.4  | 17.5  | 17.3  | 16.2  | 14.9  | 13.7  |  |  |
|                                                                        |             | MCDBR  | 14.8  | 14.4  | 14.8  | 15.4  | 15.6  | 15.6  | 17.2  | 18.9  | 18.3  | 17.4  | 16.4  | 15.7  |  |  |
|                                                                        | 5% WB       | MCWBR  | 5.2   | 4.6   | 4.9   | 6.0   | 6.7   | 7.1   | 8.1   | 8.9   | 7.9   | 7.5   | 6.6   | 6.2   |  |  |
|                                                                        |             | MCWBR  | 11.3  | 10.8  | 10.8  | 12.2  | 12.5  | 13.5  | 14.2  | 16.0  | 15.3  | 14.6  | 13.5  | 12.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.385 | 0.381 | 0.360 | 0.345 | 0.329 | 0.316 | 0.308 | 0.324 | 0.363 | 0.385 | 0.391 | 0.404 |  |  |
|                                                                        | taud        |        | 2.493 | 2.499 | 2.538 | 2.530 | 2.511 | 2.530 | 2.534 | 2.463 | 2.371 | 2.332 | 2.367 | 2.378 |  |  |
|                                                                        | Ebn at Noon |        | 960   | 951   | 947   | 921   | 898   | 893   | 913   | 929   | 929   | 936   | 947   | 941   |  |  |
|                                                                        | Edh at Noon |        | 117   | 114   | 105   | 99    | 93    | 87    | 89    | 103   | 122   | 133   | 131   | 131   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.01  | 6.99  | 6.53  | 5.90  | 4.91  | 4.45  | 4.87  | 5.81  | 6.79  | 7.41  | 7.55  | 7.35  |  |  |
|                                                                        | RadStd      |        | 0.61  | 0.42  | 0.51  | 0.44  | 0.26  | 0.33  | 0.23  | 0.24  | 0.39  | 0.43  | 0.40  | 0.52  |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (3 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page